Inequalities — problem 7.49

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The polynomial $ax^2 + bx + c$ with $a > 0$ has real roots x_1, x_2. Prove that $|x_i| \leq 1$ $(i = 1, 2)$ exactly if $a + b + c \geq 0$, $a - b + c \geq 0$, $a - c \geq 0$.